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Highlights

- A Grounded Theory study in context of agile transformation process
- Causes and consequences of inadequate transition of organizational roles
- Factors influencing transition of organizational roles from traditional to agile

Transition of organizational roles in Agile Transformation Process: A Grounded Theory approach

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Abstract

This study aims to identify how traditional organizational roles are transformed towards Agile roles. A Grounded Theory study with 5 software teams in a large software development company in Spain was conducted. We interviewed 21 people, participated in Agile Coaching sessions and observed daily activities in the company during a six-month period. The transition of organizational roles during the Agile transformation process is influenced by many circumstances such as: higher management commitment and support, knowledge about Agile, project team size and culture. The theory demonstrates how enterprise factors influence the transition of organizational roles and how practitioners cope with the challenges during the Agile transformation process. Based on our results, we proposed recommendations and guidelines for organizational roles transition during the Agile method adoption.

Key words

Agile transformation process, organizational roles, software development, grounded theory

1. Introduction

Since the emergence of Agile software development methods in the early 1990s, the Information Technology companies have had an increasing interest in applying Agile framework. Many organizations claim to be Agile or have interest to use Agile methods (Laanti, Salo, and Abrahamsson 2011). The use of Agile software development has caused a significant change in the way how projects are organized and executed (Dingsoyr et al. 2012; Stettina and Hörz 2014). Due to its optimization effect and reduction of overhead typically related to the traditional software development approach, Agile methods are usually well perceived by management and developers (Qumer and Henderson-Sellers 2008).

Implementation of Agile methods helps software development companies to tailor their services and products, thus providing the ability to adapt to current dynamics in the market (Cao et al. 2009; Highsmith and Cockburn 2001; Barry Boehm and Turner 2003; Lindvall et al. 2004). Agile practices can help mature organizations to become more flexible (Lindvall et al. 2004b). Agile methods are considered suitable for small and medium-sized companies (Nguyen 2015) and projects (Barry Boehm and Turner 2003), and software development organizations are increasingly recognizing the need for agility in nearly every project they execute (Cao et al. 2009). The challenge lies in not only successfully adopting Agile practices at a project level but in efficient integration in the project environment at the organizational level (Lindvall et al. 2004b).

Agile outside of individual projects and teams, and implementation of Agile in broader organizational context - above the project level should be investigated further (Laanti, Salo, and Abrahamsson 2011; Stettina and Hörz 2014; Nguyen 2015). Addressing people

conflicts is vital to adoption and incorporation of Agile practices in organizational processes (“Management Challenges to Implementing Agile Processes in Traditional Software Development Organizations” 2005).

The research objective of this paper was to investigate the Agile transformation process in a large IT Company, and to identify the key challenges arising, and strategies for solving them in the Company. Most studies focus on the outcome of the transition process and the desired state of the organization after the transformation, but the exact steps of the transformation are not defined in detail and depend on each particular situation. Also, organizational roles transformation during the Agile method adoption was not identified in the research literature. Therefore, in this research the evolution of the Scrum method adoption in the organization was explored. Parameters influencing this process and possible solutions used in the company are presented. In order to fulfil the research objective, the grounded theory method was chosen to be applied in the company where the research was conducted. The resulting theory was organized about the core category named “Inadequate transition of organizational roles”.

The Company where the research was conducted has 350 employees, with the main operations located in Spain and with more than 3000 clients in the tourism sector. The main product of the company is the Hotel Distribution platform. The previous process model used in the company was most similar to waterfall (plan-driven) methodology. The large - scale Agile transformation process in the company started at the end of the year 2014. It was implemented in phases in the IT department with the aim of global transformation of the entire Department, where 190 employees were working. The transformation began with different groups of project teams, resulting in the adoption of the Scrum method in almost all the project teams working in the IT department. Since the teams were collaborating on a large product – Distribution platform, the change process is considered as a large-scale transformation (Dikert, Paasivaara, and Lassenius 2016). The educational cycle of the employees started at the end of 2014, and by the time the researchers initiated the data collection process, the Scrum method was in use and implemented in most of the teams in the company IT department. Data was gathered through interviews and participation of the researchers in observation and Agile Coaching activities. Agile Coach and 20 employees from five project teams were involved in the semi-structured interviews, totalling with 21 persons interviewed. Data gathering, analysis and theory forming were performed following the standard grounded theory (GT) proposed by Glaser (B. G. Glaser and Strauss 1967), and additionally the specific consideration for conducting and reporting GT was embedded in the study (Stol, Ralph, and Fitzgerald 2016).

This research study presents the following contributions: (1) description of the organizational roles transition during the Agile transformation process, (2) the application of the grounded theory method in software engineering, (3) implications of the results on theory and practice.

The article is organized as follows: The next session provides a short review of related literature. Section 3 provides detailed steps of how Grounded Theory was applied as a research method in this study. Section 4 shows the results obtained in this study. This is followed by a detailed discussion in section 5 where the obtained results and relation to relevant literature are discussed. Section 6 presents the evaluation process of the results, followed by section 7 where limitations of the research are defined. Finally, the last section 8 concludes the paper and discusses possible work in the future.

2. Related work

In this section other studies related to the research are described briefly. The number of organizations adopting Agile practices is significantly increasing (Soundararajan and Arthur 2011). The context of this research was to observe a Software Company moving to Agile. In the literature different frameworks facilitating Agile transformation may be found. Different approaches to Agile transformation are available such as a large number of academics' and practitioners' research reports on the topics of Agile process tailoring or specific Agile method adoption. Challenges and obstacles of the Agile methods adoption was also a significant area of the research.

Different research objectives under the context of Agile transformation process may be found in the research literature: challenges and obstacles during transformation process (Sidky, Arthur, and Bohner 2007; Chan and Thong 2009; Lindvall et al. 2004b), Agile framework development (Soundararajan and Arthur 2011; Asif Qumer and Henderson-Sellers 2008; Cao et al. 2009), balancing the coexistence of Agile and standard methods (Campanelli and Parreiras 2015) or assessment of achieved transformation (Chuang, Luor, and Lu 2014). People conflicts present an important issue to be managed in the in change management process (Wu and Tang 2015). Different strategies for resolving people conflicts may be used: understanding how communication occurs, educating stakeholders, transmitting the software issues to be understandable to customer and management, emphasizing value, choosing the right people and awarding for results and reorienting reward system (B. Boehm and Turner 2005). Research results of Rohunen imply that there is no study that clearly specifies how to adopt Agile practices and further investigation of Agile adoption framework and strategies, specifically in large companies, is needed (Rohunen et al. 2010).

Many authors report on challenges and issues encountered during the Agile transformation process (Nerur, Mahapatra, and Mangalaraj 2005; B. Boehm and Turner 2005; Lindvall et al. 2004b; Stettina and Hörz 2014). Management faces several barriers when adopting Agile approach in the traditional organization (B. Boehm and Turner 2005). The iterative nature of Agile methods introduces new challenges to the current management practice (Stettina and Hörz 2014). A group of authors identified four groups of challenges during the Agile method adoption: management and organizational, people, process and technology (Nerur, Mahapatra, and Mangalaraj 2005). They further defined challenges specific for management and organizational aspects: organizational culture, management style, organizational form, management and/or development knowledge and reward system. A group of authors conducted a multi case study in three organizations that had adopted extreme programming practices and they identified as the main challenges to adopting Agile methods: development process related, customer related, developer related and organization/management related (Cao et al. 2009). The complexity of adopting Agile practices is high because of resistance to change, necessity of higher management involvement and organizational culture (Campanelli and Parreiras 2015; Cao et al. 2009; Dyba and Dingsoyr 2009). A group of authors conducted research in four large companies adopting Agile techniques and identified the following incompatibilities between the pilot project team and general organizational setup: customers, requirements, architecture, legacy systems, other teams, quality systems, change control boards and organizational software processes (Lindvall et al. 2004b).

Most of the difficulties in adopting Agile practices come from the project surrounding, not from implementing Agile practices in the project itself (Lindvall et al. 2004a). Higher management support and commitment was identified as one of the causes for adequate evolution of organizational roles. Management style in the traditional setup is command and control, while in Agile the focus is on leadership and collaboration (Nerur, Mahapatra,

and Mangalaraj 2005). Large organizations often follow defined software processes (Mesquida et al. 2014; Sidky, Arthur, and Bohner 2007; Mesquida and Mas 2015) and applying new Agile practices at the project level might bring double work (Lindvall et al. 2004b). Interplay of practices across functional roles should be better understood in order to comprehend organizational agility (Stettina and Hörz 2014). Larger software development organizations tend to have more organizational roles, while smaller companies tend to merge roles in a manner to create the best fit to their operations limitation (Yilmaz, O'Connor, and Clarke 2015). People behaviour is an important factor in the organizational change (Vedenik and Leber 2015). Four specific challenges were identified for people conflicts: management attitude, logistical issues, handling successful pilots and change management ("Management Challenges to Implementing Agile Processes in Traditional Software Development Organizations" 2005).

Contemporary Agile methods are not anti-method and Agile adoption process requires just as disciplined approach and tailoring as any other traditional model (Fitzgerald, Hartnett, and Conboy 2006). The importance of well adapting Agile methods is well known, but exact steps and general adoption process are described only to a certain extent (Cao et al. 2009; Fitzgerald, Hartnett, and Conboy 2006). Many experience reports on Agile method adoption are available, but there is no unique, widely accepted solution on how an organization should conduct Agile transformation. Agile transformation process is often described as a cultural change of software development (Laanti, Salo, and Abrahamsson 2011).

Different factors influencing Agile adoption can be divided into four groups: ability related, motivation related, opportunity related and Agile methodology characteristics (Chan and Thong 2009). The way of introducing the Agile process in a company is very important, the process is useful in practice for successful Agile adoption (Cohn and Ford 2003). Agile methods are useful in practice once they are carefully personalised and adapted to the company (Laanti, Salo, and Abrahamsson 2011). Integrating new practices with the existing processes in organizations requires process tailoring (Lindvall et al. 2004b; Fitzgerald, Hartnett, and Conboy 2006).

Agile principles, values and practices are well known by software industry, but the problem appears when they need to be applied in practice (Rohunen et al. 2010). Lack of complete strategies and guidelines for Agile adoption makes this endeavour harder to accomplish. The Agile adoption process strongly relies on organizational environment. When adopting Agile methods, large software development organizations have to deal with organizational transformation towards Agile and they commonly have standardized processes and organizational structure not suitable for merging with Agile software development (Paasivaara and Lassenius 2014). Agile practices and methods often have to be tailored to be incorporated into the existing company processes (Rohunen et al. 2010; Campanelli and Parreiras 2015; Dyba and Dingsoyr 2009).

Different frameworks of Agile adoption may be found in research literature. Objective, Principles and Practices (OPP) framework is a comprehensive approach to assessing the "goodness" of particular Agile method for a specific company based on adequacy, capability and effectiveness (Soundararajan, James, and Balci 2011; Soundararajan and Arthur 2011). Agile software solution framework (ASSF) is developed to assist management in evaluating the level of agility they require and how to identify suitable manner to introduce the agility into their company (Qumer and Henderson-Sellers 2008). A group of authors identified the absence of guidance and assistance in the Agile transformation process in the scientific literature, and in response to that created a framework to be used as a structured approach towards successful adoption of Agile practices (Sidky, Arthur, and Bohner 2007). The framework consists of two main components: Sidky Agile Measurement Index (SAMI) for estimating Agile potential, and a

set of four-staged processes with a measurement index used to determine which Agile practices may and should be integrated with the organisation.

3. Methodology

This empirical study was conducted based on the classic grounded theory proposed by Glaser and Strauss (B. G. Glaser and Strauss 1967). GT is a research method where theory is generated from the obtained data. This research method produces a set of integrated conceptual hypotheses systematically generated to inductively develop a theory about substantive area (B. G. Glaser and Holton 2004). It is a set of carefully grounded concepts organized around a core category and integrated into a hypothesis.

Grounded theory is one of the best research methods to answer the questions such as “What is going on in the area?” and to generate substantive theory. The research question is formulated during the research process of GT, and not as a result of an extensive literature review done in advance, prior to initiation of research (B. Glaser 2008). The GT study as a result brings the aspect of actors involved in the process of interest.

When there is no clear hypothesis or research problem defined upfront, GT is a suitable method. If some area has not been researched in detail before, or obtaining a new fresh perspective is intended, GT brings good results in creating a new theory. The GT approach proved efficient in the investigation of the Agile method adoption in the Company, resulting in a well-integrated and saturated theory.

Interview participants were chosen based on the company’s organizational structure, development teams experience in practicing Agile, and suggestions from the higher management. The IT department in the Company is divided into different sub-departments related to the final product (Hotel Distribution platform): integrations, platform, booking engine, client reservation management and revenue management. The five teams chosen to participate in the research participated in the first cycle of the education process. The whole IT department is organized in five sub-departments, and the selection of one team from each sub-department provided the research with a diversified sample representing the IT department as a whole. Moreover, due to the fact that the sub-departments collaborate and work jointly on the final product, it was important to involve teams from all the departments in the research. Also, an important prerequisite was that the employees participated in the research on the voluntary basis and the initial plan was not to be changed due to this fact.

In Figure 1 a graphical representation of the research process followed is shown.

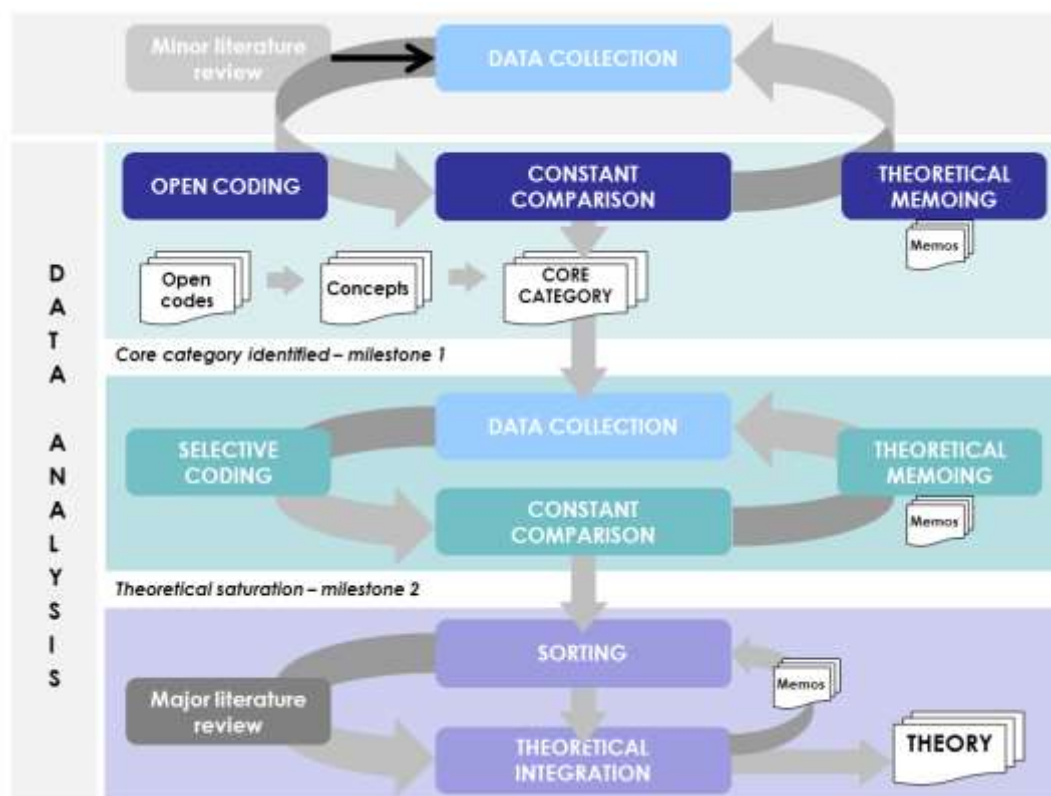


Figure 1. Research method []

3.1. Data collection

Data collection is the first phase in the GT method. It is repeated all the time through the research process up to the point of data saturation. When no new information is acquired through the data collection process, and new concepts are not identified, researchers should terminate the data collection process (B. G. Glaser and Holton 2004). Rich data provides participants' opinions, feelings, actions, as well as the context of their interaction with others. GT, as a method, requires constant data collection and analysis activities. Right from the beginning, the focus of the researcher should be on the conceptualization while using a comparative analysis method (B. G. Glaser and Strauss 1967). Basically, analysis of data should direct data collection to be conducted, and the researcher should constantly search for concepts, analyse data and return a step back to revise concepts (B. G. Glaser and Holton 2004). After the most important concerns and relations of the participants are identified, first concepts are created. Following that, researchers should proceed with theoretical sampling and, based on the first categories identified, they should decide which type of data should be further collected.

In this research, we used semi-structured interviews with an open ended questionnaire to gather primary data. Participation in coaching meetings and observation activities enabled secondary data collection. Three researchers were involved in the data collection process, while the fourth was involved in data analysis. The principal researcher participated in all

the data collection activities, while the other two researchers' involvement depended on the type of activities. Minimally two researchers participated in the interviews conducted. All the data collected were reviewed separately by two researchers, and differences in findings were discussed among all four of them. Having four researchers to participate in this research was important for objective data collection, conceptualization and analysis.

Two meetings were held prior to the initiation of the data collection activities. During the first kick-off meeting, higher management, team leaders and researchers agreed that participation of the employees in the interviews should be on a voluntary basis. The second meeting was held with the team leaders, Agile Coach and the researchers, where the time schedule of data collection was established. It was agreed to perform data collection activities during a six-month period (from March to September 2015) as an initial data collection phase. The direct effort of employees was estimated to the maximum of 30 hours during the six-month gathering data period in the company. After that, if needed, a second round of data collection would have been conducted in the future. The sociologist trying to discover theory cannot define in the beginning of the research how many groups he will sample during the entire study, rather he can only sum up the number of interviewees in the end (B. G. Glaser and Strauss 1967).

After the initial meetings, the team leaders presented the planned research project to their team members, who later on agreed voluntarily to participate and share their experience about the Agile transformation process in the company. The employees responded very willingly to our request and more than 20 employees were willing to participate. Agile Coach was not a Company employee, but he was involved in the Agile transformation process. Study participants varied in their technical and Agile experience, age and organizational roles. They had from 1 to 18 years of experience in software development, and from 1 to 8 years of Agile experience. Most of the team members were involved in creation and software delivery. They had different organizational roles before the Agile method adoption, namely: developer, business analyst, tester, and project, program and product manager.

Interviews with the employees were planned to be conducted separately from other data collection activities - observation. The time schedule of the interviews was created in a manner to have enough time and to attain the right pace to analyse and conceptualize the obtained data. Interviews lasted for one hour on average. In most of the cases only one or two interviews were conducted in a single day, and a maximum of four in the same month. 21 interviews were conducted in this research. Table 1 shows the details of 21 interview participants, grouped by the teams they belong to. The roles of the interviewees before Agile adoption (in the previous process model) were: System Analyst (SA), Requirement Engineer/Functional Analyst (RE), Software Architect (SA), Software Developer (SD) and Project Manager (PM). Organizational roles were transitioned in the organization towards Scrum roles: Product Owner (PO), Scrum Master (SM), Team Member (TM).

Table 1. Participant teams and members

Team	Team members	Age mean	Participant	Role before Agile adoption	Agile role	Software development experience (years)	Agile experience (years)	Years in company
T1	5	31.2	P1	SA	PO	15	4	10
			P2	SD	SM	5	3	3.5
			P3	RE	TM	7	3	3.5
			P4	SD	TM	8	2	2.5

Team	Team members	Age mean	Participant	Role before Agile adoption	Agile role	Software development experience (years)	Agile experience (years)	Years in company
T2	7	31.7	P5	SD	TM	5	1	2.5
			P6	SA	PO	9	3	7
			P7	SA	SM	7	3	6
			P8	SD	SM	4	2	1
			P9	SD	TM	3	1	1
T3	10	33.3	P10	PM	PO	15	2	12
			P11	SD	SM	5	3	1
			P12	RE	TM	7	1	4
			P13	RE	TM	9	1	6
T4	10	34.5	P14	PM	PO	15	5	10
			P15	RE	SM	14	5	10
			P16	SA	TM	10	3	7
			P17	BA	TM	12	3	1
T5	16	32.3	P18	PM	PO	18	2	10
			P19	RE	SM	12	2	10
			P20	SA	TM	8	2	5
			P21	-	AC	16	8	-

The open ended questionnaire was adopted from (Paasivaara & Lassenius, 2014). The questionnaire was modified in line with the current situation in the company where the research was conducted. The questionnaire A.1 from the Appendix was used in the research. In the beginning, some questions that were not applicable to the company were removed, namely in the sections two, three and five of the questionnaire. In total, eight questions were removed in the beginning and some terms and expressions were modified throughout the whole questionnaire to better align to the terms and methods used in the company. The final version used in the survey is attached in the Appendix. For instance, questions: “How many persons do you have at the moment at each site? How many teams at each site? What are the team sizes at each site?” were not applicable since the company does not have multi sites and, therefore, they were removed from the questionnaire.

The interviews started with general questions about relevant experience of the interviewees and general implications of the Agile transformation process, followed by questions about organizational structure and roles before and after the transformation. Later on, followed questions about coaching and team level practices, possible improvements, new tools used in daily activities, scaling Agile and cross team practices, quality indicators, product management and general questions about the Agile transformation process.

During the interviews, when the core category was identified, some questions were added while others were removed for the second phase of interviews. The questions that were removed are 2i), 4c), 5b), 8b) and the questions that were added are 3g), 3h), 3i), 3j). As an additional change in line with the identification of the core category, the researchers made sure to spend more time with questions in section 3 during the second phase of data gathering activities (12 minutes as a minimum).

Before the start of each interview, the interviewees gave their consent to have the sound of the whole conversation digitally recorded and to the creation of transcripts and

conclusions for scientific purposes. During the interviews, which were conducted in English, the subjects were guided to discover both successful and less successful projects in which the team members were involved, and to unveil the wider aspect of the Agile transformation process. The participants had the opportunity to share their concerns, describe changes, improvements and drawbacks, and also to offer possible strategies for the arising issues, during and after the transformation process. The transcribed data was constantly analysed, compared and reviewed in researcher pairs as proposed by the GT method (B. G. Glaser and Strauss 1967). After each interview, the researchers created transcripts which were sent directly to the interviewees to obtain their feedback and potential clarification where needed. After that, the interviewees sent their feedback on transcripts by email. The responses from the interviewees were further discussed in direct contact with the researchers, during and after the coaching sessions, as well as during the final meeting held after the end of the interviewing process, when the initial results were presented by the researchers to all the participants (five project teams involved in the research study).

After the interviewing process was finished (in December 2015), the final meeting was held to present the initial results from our study to a wider group of employees in the company (all the team members from five project teams involved in the research study). The intention was not to confirm if the presented theory demonstrated their exact opinion, rather it was towards obtaining an additional point of view in terms of sorting and writing up the theory.

Apart from the interviews, the researchers participated in observation activities. Observation activities lasted 25 hours in total and were not accounted as a direct effort of the employees.

Two researchers observed 5 Agile Coaching sessions in the company that were organized as a follow-up of training sessions provided by external consultancy. In these sessions, the Agile Coach would visit all the teams in front of the Kanban board and discuss different issues and potential suggestions and innovations. Coaching was delivered by an external Agile Coach, and the researchers were observing the sessions with the opportunity to ask for further details from the participants involved, but without obligation of delivering specific content or coaching the participants. Observation in the coaching activities was a valuable opportunity to gather data and observe the evolution of team behaviour and incremental improvements on a monthly basis and to discover the group thinking of all the team members in addition to the individual interviews. Each coaching session lasted 2 hours, meaning that the researchers in total observed 10 hours of the coaching sessions.

The researchers also gathered data through another 15 hours of observation activities of regular daily activities of the team members. The researchers created a schedule of observational activities depending on the daily schedule of each team, with the intention to participate in the team meetings when possible, and to observe the data created by the teams. Each team established specific daily routines and our intention was to adapt observation activities to the team routines (dynamics). The researchers participated in two sprint planning meetings, two review meetings, one retrospective meeting and five daily meetings. We participated in two meetings of each team: one daily meeting and one other meeting (retrospective, review or planning). Also, available documents were reviewed, such as meeting minutes, Kanban board with post-its, burn-down charts, task board in Jira, user stories and use cases. Documents were analysed in the company and clarification of details was provided by the team members on site when needed. Later on, the researchers analysed the documentation gathered, which was further related to other results obtained in the interviews as a part of memoing activities and structuring/conceptualizing the obtained data. Therefore, the researchers were taking notes during the observation meetings and the

obtained data were primarily used to enrich the codes, concepts and categories and relations among them through memoing activities.

The initial plan of six months' data collection was successfully fulfilled. Data collection activities within this period were changed and adopted due to the specific team dynamics and project necessities, but the overall duration was not different from planned. Most of the changes in the planned schedule were due to seasonal holidays and data analysis rhythm. Due to the seasonal holidays in August and unexpected workload at some point of time, some interviews were rescheduled and conducted in two different weeks in the same month. The overall plan to have three or four interviews per month was maintained throughout the whole data gathering period. For instance, in August, interviews were performed after observation activities, resulting in three separate interviews during that month, but in the other months of data collection activities, interviews were held bimonthly (every two weeks).

The planned pace was well estimated having in mind the need for constant comparison, memoing and conceptualization as a prerequisite of the GT method. So, it was of high importance to have time to analyse data before continuing with data collecting activities. The researchers were present on site every week, since interviews, coaching and observation activities were planned to be in different weeks. On average, during the six-month period of the data gathering process, the researchers spent 8 hours per month on site to collect data. In total, the researchers participated in interviews with 21 participants summing up to 21 hours and 48 minutes, observation of five coaching sessions summing up to 10 hours and observation of a team's daily activities - ten team meetings and reviewing documentation, summing up to 15 hours.

3.2. Data analysis

GT's data analysis, also called data coding, starts right after first data is collected and it is carried up to the point of emergent theory creation (B. G. Glaser and Strauss 1967). The conceptualization of data through coding is the essential step in GT development (B. G. Glaser and Holton 2004). Substantive code represents the essential relationship between data and theory. Coding gets the researcher of the empirical level by breaking the data, followed by conceptual regrouping of the data into codes becoming theory, thus explaining the relations among the data. Substantive codes conceptualize the empirical substance of the research field and theoretical codes conceptualize how substantive codes are related to one another as a hypothesis further integrated as a cohesive whole during the process of theory creation (B. G. Glaser and Holton 2004). In the research, a multi-level coding process of data analysis is used as explained in the following sections.

Data analysis was divided into three phases:

1. Open coding resulting with core category identified (section 3.2.1)
2. Selective coding terminated when data saturation is identified (section 3.2.2)
3. Sorting and theoretical integration resulting in theory write-up (section 3.2.3)

3.2.1 Open coding, constant comparison and theoretical memoing

The first phase of data analysis, performed right after initial data has been collected, is named open coding, also referred to as substantive coding. Researchers should use their ability to generate codes and find relevance with the open coding method (B. G. Glaser and Holton 2004). Open coding begins with line-by-line or sentence-by-sentence coding of the

transcribed data. Word-by-word data is time-consuming, and it can be confusing and tedious, therefore the researchers decided to follow the line-by-line method. Line-by-line coding forces the researcher to saturate and verify the emerged concepts (B. G. Glaser and Holton 2004). Identifying and focusing on key points, instead of words, was found suitable in this study.

On a monthly basis, the researchers were able to conduct and participate in: three or four interviews of one hour per interview on average, two hours of observation and two hours of coaching activities. 14 interviews were conducted in the open coding phase. The core category was identified after four months, at the moment when 14 interviews had been conducted. In accordance with the GT method, after the core category identification, we continued with data gathering analysis until data saturation, which in our concrete case resulted in 7 interviews after the moment of the core category identification. Data were collected in different weeks during the same month providing the research with the optimum pace to gather data and analyse the data through memoing and constant comparison. Explicit primary data source can be chosen, and theoretically sampled from the remaining data based on the leads arising from the primary data source (Stol, Ralph, and Fitzgerald 2016). In this research interviews were conducted, and input from the observation activities was used to enrich the leads from the interviews throughout the whole process of data gathering and analysis.

The first data collection activities started with the conduction of the first two interviews in March 2015. Immediately after conducting the first round of interviews, the researchers started with line-by-line coding, leading to the first open codes in the study. Through open coding, key points were unveiled and codes were assigned to them. Therefore, the data obtained from the first interviews were analysed and first open codes were assigned to the interview transcripts. Also, first memos were created from the interviews, observation and coaching activities. Therefore, an example of the open coding process is presented.

Step 1. Open codes are generated from collected data

Examples of open codes are presented in Table 2.

Table 2. Open coding example

Category	Concept	Open code	Statement	Source	Role	Interviewee
Strong team culture hinders changes	Team size	Team size	"When a team is big, it is harder to divide tasks and be Agile..."	Interview w	Team member	P16
			"Big teams need two Product Owners"	Interview w	Scrum Master	P15
			"A small team is Agile by default"	Interview w	Scrum Master	P11
	Management style	Collaboration versus delegation	"We would like that everybody can take his/her task, not to be delegated. Someone just likes to delegate instead of collaborate"	Interview w	Product Owner	P6
		Traditional management	"I feel that it is still a challenge to change the mindset of people from management. They still want to send some task from an important client and to be done for yesterday"	Interview w	Product Owner	P1
	Multiple	Scrum	"I as a Scrum Master also talk	Interview	Scrum	P15

Category	Concept	Open code	Statement	Source	Role	Interview ee
	roles	Master role	to the client and take little work load of our Product Owner in prioritizing tasks, and asking the client for more details about requirements”	w	Master	
		Product Owner role	“Basically our Product Owner is like a Portfolio Manager. I also need to work on priorities and backlog grooming...”	Interview w	Scrum Master	P15
	Team culture	Team history	“Clients are used to contacting me from before...”	Interview w	Scrum Master	P15
		Team experience	“When a team works less time together, it is much easier to bring changes and new technologies”	Interview w	Scrum Master	P11
		Team processes	“Anyhow, I would not call this a transformation but more adding valuable tools and techniques to our processes...”	Interview w	Product Owner	P18
	Resistance to change	Resistance to change	“It is harder to implement Agile in the team that works long together”	Interview w	Scrum Master	P19
	Splitting team	Splitting team	“We wanted to split our team in two”	Interview w	Scrum Master	P11
	Double work	Double work	“The Product Owner and the Scrum Master would attend all the meetings from one and other group and this was recognized as double work”	Interview w	Scrum Master	P11
	Team maturity	Team maturity	“If this Agile change had happened 15 years ago, when we were established as a group, it would have had much greater impact on us”	Interview w	Product Owner	P18
		Innovative environment	Team members are open to changes. Retrospective meetings are used for improvements. Everybody is participating.	Observation	-	Researcher
		Group decision making	“The team is democratized. We are all involved in decision-making. Not everybody can do everything, but we have improved”	Interview w	Product Owner	P10
		Shared responsibility	“The team has to understand that work to be done is a group responsibility, not individual”	Observation	-	researcher
Lack of knowledge about Agile	Agile training	Agile practice	“When we began using Agile, we were missing practice”	Interview w	Team member	P5
		Practical exercises	“We needed more detail with some practical exercises”	Interview w	Team member	P12
	Agile	Need of	“A course without coaching has	Interview	Team	P17

Category	Concept	Open code	Statement	Source	Role	Interviewee
	Coaching	coaching sessions	low value”	w	member	
			“Coaching sessions are very useful. We identified techniques to be customized in our project”	Interview w	Scrum Master	P2
		Coaching in meetings	“The Agile Coach should participate once in all types of meetings (planning, daily and retro)”	Interview w	Scrum Master	P11
	Agile roles	Knowledge about Agile roles and practices	“Scrum Master was not sure how to resolve impediments related to other teams”	Observation		researcher
	Communities of practice	Agile guru	“We are missing a responsible person for Agile in the company, an Agile guru”	Interview w	Scrum Master	P15
		Kaizen team	Scrum Masters are planning to make an Kaizen department/team for continuous improvement	Observation	-	researcher

Step 2. Open codes are grouped into concepts

Concept is the naming of an emergent social pattern grounded in research data (B. Glaser 2008). New concepts surface and new observations fit into existing concepts (B. G. Glaser and Holton 2004). More indicators were developed through repetitive reading of the transcripts, and each key point was compared to the previous key points. Constant comparison process is a fundamental concept of GT and it is embedded in this study (B. G. Glaser and Strauss 1967). New concepts were compared to the previously emerged concepts and codes, and the researchers always went through both the current and the previous transcripts in order to identify new concepts, or modify the existing ones.

Following with the example in Step 1, from the above open codes the concept *Team culture* was obtained. During the open coding process different concepts were identified: *Higher management commitment and support, team size, team culture, knowledge about Agile, Product Owner proxy, product complexity, technical expertise, organizational roles, human aspect*, etc. Through the constant comparison method open codes were grouped in the emerged concepts, and after a few iterations of reorganization and concept fine tuning, categories were identified.

After conducting each interview, the researchers created memos focusing on the identified concepts, thus providing more data to be used for further analysis. Memos are theoretical notes about data and relations between categories such as ideas, feelings, thoughts and concerns related to the concept (B. G. Glaser and Holton 2004). Glaser believes that theoretical memoing is an essential stage for GT (B. G. Glaser and Strauss 1967). Memoing is not optional, and by following this technique researchers make certain to decrease the data gathering pace, and consequently not to draw conclusions on the

theoretical framework and core concepts in the premature stage of the research (B. G. Glaser and Holton 2004). Memos proved very helpful in the creation of relationships between the concepts and categories, and in saturating codes through theoretical coding. Theoretical memoing was done throughout the whole data analysis process, memos were created from the beginning of data analysis, and in the final phase (theoretical coding) they were integrated into theory.

To provide an example of the constant comparison and theoretical memoing, we can refer to two of the codes identified in the research: *Agile Coach* and *Agile guru*. On the one hand, *Agile Coach* was the open code identified during the first month of the data collection process, mentioned more times during the interview process. On the other hand, *Agile guru* was identified in the fourth month of the interviews. The researchers were comparing the new code the *Agile guru* to the previously identified open code the *Agile Coach*. As a result, the following memo was created: “*Agile Coach is an internal employee in the company, from higher management, who knows well the Agile methods and supports the Agile transformation, like an Agile sponsor in the company. Agile Coach, in this case, was an external resource temporarily involved in the education and coaching activities during the transformation process. After the education and coaching sessions are finished, it would be important to have a person in the company with high authority (Agile guru), who would be available for occasional questions by different teams regarding the Agile method adoption.*”

At first glance *Agile guru* seemed to be the same code as *Agile Coach*, but after reviewing the transcripts, the presented memo was created and *Agile guru* was recognized as a separate open code.

Step 3. Category emerges from a group of concepts. Finally, one of the identified categories is the core category.

The primary function of the core category is to integrate theory and it can be any type of theoretical code that represents some condition, consequence, process etc. (B. G. Glaser and Strauss 1967). Core category is at the higher hierarchy level of concepts, and iterative comparison of concepts helped to find categories. Core category is accounted for the most of the variation around a research problem or a concern that is in the objective of the investigation conducted (B. G. Glaser and Holton 2004). Each category represents a group of concepts that are associated with the same phenomena. Throughout the constant comparison process, new insights were recorded as memos, which later helped in creating new categories, their properties and dimensions. Different categories were identified during the open coding process, but finally the core category was identified.

The core category was identified in the fourth month of the data collection process. At the moment of identifying the core category 14 interviews and 16 hours of observation activities had been conducted. Other categories identified at that point of the data gathering and analysis process were also considered to become the core category. Three potential core categories emerged from the open coding process: Lack of knowledge about Agile, Strong team culture hinders changes and Inadequate transition of organizational roles. The core category should have the most explanatory power (B. G. Glaser and Strauss 1967), clearly present the problem that is the focus of the study (B. G. Glaser and Holton 2004) and it should encapsulate and explain the grounded theory as a whole. The interviewing strategy changed after the core category was identified. Four questions were added in the interviews and more time was devoted to questions related to this core category.

The core category, *Inadequate transition of organizational roles*, was identified in the fourth month of the data collection process. Three potential core categories emerged from the open coding process: *Lack of knowledge about Agile*, *Strong team culture hinders changes* and *Inadequate transition of organizational roles*. The core category should have

the most explanatory power (B. G. Glaser and Strauss 1967), clearly present the problem that is the focus of the study (B. G. Glaser and Holton 2004) and it should encapsulate and explain the grounded theory as a whole. *Lack of knowledge about Agile* was the first one to be identified and could not encapsulate the theory as a whole. Further on, the *Strong team culture hinders changes* category is related (grounded) to many codes in the research conducted, and the research conducted showed that it was an important factor in the context of software companies moving to Agile. On the contrary, the *Strong team culture hinders changes* category failed to encapsulate some important aspects and generate a theory as a cohesive whole. *Strong team culture hinders changes* relates strongly to the project level, and higher organizational levels and aspects such as multi-project environment, as well as higher management commitment and support, would have been left out from the resulting theory. The core category identified, *Inadequate transition of organization roles*, had a well developed range of properties and dimensions, and it was the best candidate to integrate the other identified categories. In other words, definition and transformation of organizational roles can be related to all the organization levels, and both to the team and individual aspects in terms of their organizational role transformation. Therefore, it was an additional reason why it was the best fit to gather the other categories in a cohesive whole, and to be discovered in this research as an important aspect in the context of the software company's adoption of the Agile methods.

The open coding process finished once the core category was identified. The researchers continued with focused data gathering in relation with the core category identified. Selective coding began when the core variable was identified and the process involved only the categories related to the identified core category.

3.2.2 Selective coding

Selective coding was performed after open coding. In this phase researchers should continue with the data collection process, but with a focus on collecting data on categories for the generation of properties and hypotheses (B. G. Glaser and Strauss 1967). Selective coding was done in the same manner as open coding but now it was more "selective" and the focus of the researcher was to obtain and analyse the data related to the categories identified in the open coding phase.

Selective coding started after the core category was identified. Therefore, at the end of the fourth month of data collection, the selective coding process was initiated. The researchers continued with 7 more interviews and 4 hours of data gathering through coaching and observation activities in the last two months of data collection. After successful identification of the core category, the research process should be aligned with the findings and data collection activities may be tuned for the selective coding activities in order to reach better theoretical saturation. Two strategic questions should be answered before conducting theoretical sampling (B. G. Glaser and Strauss 1967): how many groups should be chosen and to what degree should one collect on the single group? In the company where the research was conducted, the researchers included five project teams in the data gathering process. During the first 14 interviews (open coding phase), the participants from each team were invited to interviews. During the last 7 interviews (selective coding phase) a plan was created to make sure that minimally one participant from each team was invited. The teams involved in the research were coming from different departments in the company and the plan on whom to interview was not changed after the identification of the core category. Team members were dealing with different functionalities of the Hotel Distribution platform (related to their departments) in the company and, in that sense, the sample was diversified and there was no need to look for

other interviewees to obtain different perspectives. Rather, the researchers decided to obtain more detailed data from the five teams, thus observing the behaviour patterns within the same teams and enriching category properties and obtaining hypotheses in that way. The questionnaire was modified and more time was devoted to the questions related to the core category.

During this iterative process, the researchers were focused on saturating the categories under development, their properties and relations of one to another. The researchers continued with creating memos as an important technique to obtain and save relations among data and codes, which proved very useful in saturating the properties of different categories. Even selective collection of data, with the aim of obtaining category properties and hypotheses tends to result in many excess data (B. G. Glaser and Strauss 1967). The researchers went back to the data from the previous interviews and observation activities in order to reorganize and saturate the categories, concepts and relations among them. It is important to perform comparative analysis and compare codes to codes or codes to categories (B. G. Glaser and Strauss 1967). The previously provided example with codes of *Agile Coach* and *Agile guru* provides a good example of constant comparison during the selective coding process.

Selective coding was terminated when theoretical saturation was identified. Categories are saturated when gathering new data: returns codes that only fit in the existing categories and these categories are sufficiently clarified in terms of their dimensions and properties (Birks and Mills 2012), no longer sparks new theoretical insights, nor reveals new properties of the core theoretical categories. Saturation is not seeing the same pattern over and over again, rather it presents a point when new codes are not yielding new properties of the pattern, thus demonstrating the conceptual density making up the body of the generated grounded theory with theoretical completeness (B. G. Glaser and Holton 2004). After the last round of the interviews in the sixth month of the data gathering process, the researchers identified theoretical saturation. The data obtained from the last two interviews did not impose creation of new categories, and new were fitting well in the existing categories which were sufficiently explained in terms of properties and dimensions. If this was not the case, the researchers would have continued with interviews and observation activities to reach the saturation. The initial agreement with the company higher management was to perform data gathering activities in parallel with coaching activities which was limited to six months. Also, at the kick-off meeting it was estimated that interviews would not take more than 30 hours of employees' time in total, meaning that another round of interviews would have been organized after September 2015 if that had been needed, but due to identification of theoretical saturation, this was not the case.

3.2.3 Developing the theory

Once theoretical saturation is achieved, researchers should continue with review, sorting and integration of previously created memos associated with the core variable, its properties and related categories (B. G. Glaser and Holton 2004). Sorting represents sorting of concepts, not data sorting, and it provides theoretical completeness. This is the final stage of data analysis in which the obtained data are sorted, merged with extensive literature review and memos are further developed, saturated and incorporated. In this third phase of data analysis, theoretical coding is performed and a theory as a result of the GT process is written up. Integration is best when it emerges, like the concepts. A theory should never be just put together (B. G. Glaser and Strauss 1967). A theory emerges from the obtained factual data and not only from extensive literature review. According to (Lee and Baskerville 2003) four types of generalizing and generalizability may be found in

Information Systems research: generalizing from data to description, generalizing from description to theory, generalizing from theory to description and generalizing from concepts to theory. In this research the resulting Theory was generalized from concepts.

At this stage the researchers did a major review of scientific literature and merged it with the obtained results from data analysis, thus creating a theory as a cohesive whole. Theoretical sorting of memos is the key of formulating the theory as a final result of GT. The researchers took all the memos generated throughout the study and tried to relate and enrich all the categories and hypotheses in the theory. Theoretical coding, establishing conceptual relations between the substantive codes results in the development of hypotheses (Stol, Ralph, and Fitzgerald 2016).

The main question in this phase was how to sort and integrate the data as a cohesive whole. Firstly, the gathered data was used to create codes, concepts and categories. Finally, this resulted in the identification of the core category. At this stage of the research, the process of advancing the obtained theory was inverted and the researchers had to look back at the data through sorting, and constant comparison enriched the proposed theory. If integration is not found, concepts should be resorted and reintegrated to fit better (B. G. Glaser and Holton 2004). Theoretical sorting provides logic for organizing the researcher analysis and a way of creating and refining the theoretical links prompting better comparison between the categories.

All the memos were printed out and reviewed by each researcher separately with a focus on fitting them in the theory obtained. After the researchers finished reviewing the separate findings, they jointly integrated and enriched the theory after reaching consensus on all the potential improvements identified. Comparison and emergence with fit and refit was used in category development (B. G. Glaser 1978). New memos were created during theoretical sorting, and the findings were brought into relation with the data obtained in the extensive literature review.

Glaser offers 18 families of theoretical codes or paradigms helping the researcher to think analytically and set relations among concepts (B. G. Glaser 1978). Coding paradigms are conceptual templates for organizing categories and presenting how they relate to one another. To organize and demonstrate the discovered categories and relations with the core category, Glaser six C's model was found appropriate to be used (B. G. Glaser 1978) in section 4 Results.

4. Results

This section describes the gathered results of the grounded theory study conducted in order to analyse the transition of organizational roles during the Agile transformation process.

Figure 2 is the illustration of the obtained results. The coding family (paradigm) 6 C's was used for presenting the results of the theoretical coding. It consists of the following instances: Context, Conditions, Causes, Consequences, Contingencies and Covariance. *Context* section describes the setting when the category occurs, *Conditions* defines the requisites for the Core category to emerge, *Causes* section defines the reason for the Core category to occur, *Consequences* section defines the effects of the occurrence of the Core category, *Contingencies* section defines strategies used to strengthen the consequences, *Covariance* section defines changes in one category and implies changes in the other category (causes and consequences).

Context is defined as follows: Software companies moving to Agile. Since the research was conducted in one company, *covariance* is not presented in the 6C's model, since it

could be subjective to relate the influence of causes to consequences without more than one company as a sample. In the following sections *Conditions*, *Causes*, *Consequences* and *Contingencies* are explained in detail. Due to limited space only the most important quotations from the interviews have been selected to demonstrate the work performed, and facilitate the understanding of the reader.

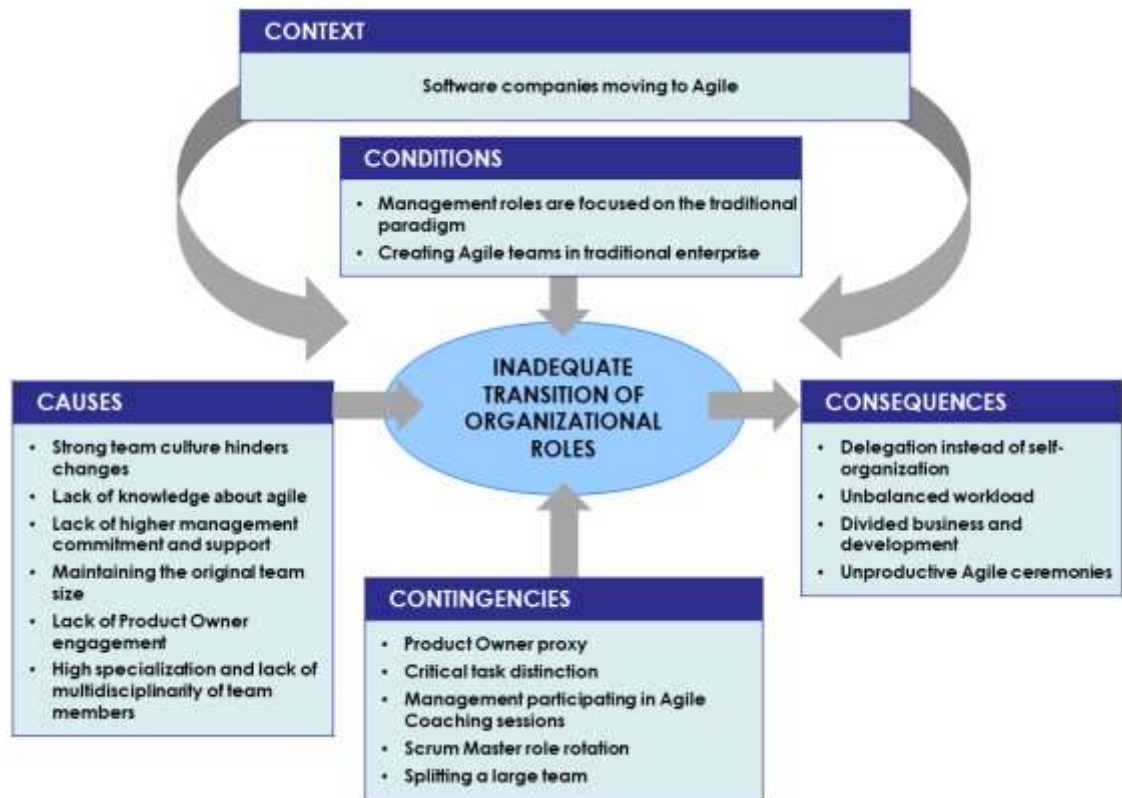


Figure 2. Theory of Inadequate transition of organizational roles demonstrated by Glaser's six C's model []

4.1 Conditions

Investigating the causes and consequences of inadequate transition of organizational roles clearly shows its occurrence pattern. Integrating new practices with the existing processes in organizations requires process tailoring (Lindvall et al. 2004b). Agile method adoption is confusing for both developers and management (Van Waardenburg and Van Vliet 2013).

Firstly, the way of planning and controlling tasks, product increments and communication within the team and with the stakeholders, have changed significantly after the Agile transformation process was conducted, and if the management roles are focused on the traditional paradigm, it is likely that causes identified in this research will take effect. The research showed that one team had problems with the management style of the Scrum Master who was still acting as a traditional project manager after the Agile transformation. Also, the reporting process was not adapted to the new situation and team members mentioned the traditional reporting as double work in the current organizational setup.

Secondly, the Scrum method was adopted at the project level and project teams began with the implementation of the Scrum practices. Changes at the higher level of organizational structures followed. Organizational roles and structure had to be adjusted, and different challenges that had arisen had to be managed in the transition period. Therefore, creating Agile teams in a traditional enterprise is the second condition for triggering identified causes in this research.

Therefore, the conditions identified in the research were as follows:

Condition 1: Management roles are focused on the traditional paradigm

Condition 2: Creating Agile teams in the traditional enterprise

4.2 Causes

Six causes influencing the Core category – Inadequate transition of organizational roles have been identified: *strong team culture hinders changes, lack of knowledge about Agile, lack of higher management commitment and support, maintaining the original team size, lack of Product Owner engagement and high specialization and lack of multidisciplinary of team members*. This section discusses each identified cause in detail.

4.2.1 Strong team culture hinders changes

Team culture was many times pointed out in different contexts by the interviewees as an important factor in the Agile transformation process. According to the interview participants, the longer the team works together, the stronger team culture is established. Interviewees mentioned different aspects of team culture such as: communication (internal and external), planning and estimation rigidity, team rules and behaviour patterns and so on. For the researchers it seemed that if the established team culture was strong before the transformation process, the adoption of the Scrum method was slower in the team. Interview participants confirmed that team members tend to do things as they know, or simply as they used to. The Scrum method defines new roles in the organization and teams with stronger culture seem to acquire these changes slower.

“When a team is working less time together, it is much easier to bring changes and new technologies, a smaller team is Agile by default, it is harder to implement Agile in the team that works long together.” - Scrum Master, P11.

Clients are used to contacting me from before... we have more projects with the same client, they cannot be separated and basically our Product Owner is like a portfolio manager and I also need to work on priorities and backlog grooming.” - Scrum Master, P15.

4.2.2 Lack of knowledge about Agile

Interviewees expressed their satisfaction with the education process prior to the transformation. Also, they expressed a positive effect of Agile Coaching activities organised as a follow-up. In Agile Coaching sessions they had the opportunity to reflect on the course content and clear doubts arising at the moment of adopting the Scrum method. Even though the interviewees were satisfied with the education and coaching sessions, the

researchers identified certain doubts and lack of knowledge about the Scrum roles and ceremonies among the interviewees. In one team there was a clear confusion about the roles. The Scrum Master stated that two Product Owners were needed to process all the requirements arriving. Having two Product Owners is not a solution; the help of a business analyst would be the adequate resolution of the problem.

The research has shown that team members had problems with understanding the exact change of organizational roles and their definition, and with the exact process of organizational roles transition. The employees also reflected on the lack of knowledge of the higher management as an additional obstacle in the Agile transformation.

“Knowledge about Agile framework in general and more importantly responsibilities are important for Agile transformation process...we had courses in two cycles not to interrupt our workflow and it was clear that others attending course later were missing the knowledge about their role... but after second educational cycle it came to equilibrium and coaching sessions helped a lot in knowledge transfer.” - team member, P2.

“Big teams need two Product Owners, me as a Scrum Master I also talk to the client and take little workload of our Product Owner in prioritizing tasks asking the client for more details about requirements.” - Scrum Master, P15.

4.2.3 Lack of higher management commitment and support

Higher management commitment and support was identified by the interviewees as an important factor influencing the Agile transformation process. It was identified in the study that the objective of the transformation process often remains unclear to team members. *Interviewees pointed out the importance of higher management commitment in educating the external Product Owners and in the facilitation of their acceptance to use Agile methods.* This fact was emphasized more by the teams with the external Product Owner. In the case when the Product Owner is not collocated with the Scrum team, or simply he or she is part of an external organization (department), then he or she is an external Product Owner.

The planning process in the company changed and it was not acceptable for the teams to break the sprint dynamics or accept the new tasks from clients in the middle of iteration.

Double reporting was another challenge to be solved in the company, as stated by the interviewees. Key performance indicators used by the management were not adapted to the new situation in the company. Team members had the feeling that the new data generated in accordance with the Scrum method was not used by the middle or higher management as it should have been. This led to double reporting and status updating in the traditional and “Agile” way.

“Way of working among team members has improved a lot, on the higher management level not that much.” - Product Owner, P6.

“One time our product manager came back from one week trip and we had a review meeting in two days but he wanted to be updated immediately, it was double work and everything was going as planned.” - Scrum Master, P11.

“I feel that it is still a challenge to change the mindset of people from management, they still want to send some task from an important client and to be done for yesterday” – Scrum Master, P8.

4.2.4 Maintaining the original team size

Five teams included in this research consisted of 5 as a minimum, and 16 as a maximum number of members. According to the interview respondents, the original size of the teams was not modified nor aligned with the Scrum method suggestions. The optimal size of Scrum teams is up to 10 members (S. Rubin 2013).

Before the Agile transformation process was initiated, the company was a weak matrix organization. Specialized resources were organized in functions and only projects involved resources from different departments. The largest project team after the transformation had 16 members who had worked jointly on one project before transformation, and they decided to maintain the number. They integrated the Scrum practices and ceremonies but did not initially think about splitting the team into two. In the interviews they said that in the beginning they thought that working together was more efficient than reorganizing since they were working on the same product. After five months the team was divided into two since Agile ceremonies were not efficient, as stated by a team member.

“We wanted to split our team in two, Product Owner and Scrum Master would attend all the meetings from one and the other group.” - Scrum Master, P11.

“When a team is big, it is harder to divide tasks and be Agile.” - team member, P16.

4.2.5 Lack of Product Owner engagement

Two main issues were identified by the interviewees regarding the Product Owner role: on the one hand, excessive interference, and on the other, lack of Product Owner engagement in Agile ceremonies and practices. The category is named lack of Product Owner engagement since in the both cases mentioned there is lack of “proper” engagement of the Product Owner.

Product Owner (PO) is the empowered central point of product leadership (S. Rubin 2013). He or she has to make sure that only the activities with the highest value are prioritized and implemented. Product Owner may be a company employee (internal) or a representative from the outside of the company (external). Each team involved in the research had a specific situation, but the researchers identified that all five teams either had an internal PO, or an external PO paired with an internal PO proxy. In the latter case, the PO proxy represented the external clients, users and other stakeholders.

From the teams involved in the research, three teams had an external PO, while the two other teams had an internal PO. Internal POs were assigned to teams whose product was mostly dependent on the other internal team inputs/requests and not directly on the external client, like in the case of the integration team whose role was to integrate the platform. External POs were assigned in the case of developing product (functionality) for a specific client. Logically, the POs were chosen by the client and they represented the central point of product management and prioritization. In the beginning of the Agile transformation process, team members decided if the external PO would be paired with a PO proxy, an employee in the company. In our case, all three teams having the external PO decided to dedicate one PO proxy from their team. Two PO proxies had a technical role before the transformation and one was from the business side. The two internal POs involved in the research had a technical background before the transformation. The interviews were conducted with the employees in the company, so two internal POs and three PO proxies (representing external POs) were interviewed. The interviewees stated typical challenges with external POs and what was expected to be resolved by PO proxies: lack of knowledge

of the PO about Agile, high workload of the external PO, POs not performing their role adequately, or complex stakeholder communication (having client and users from different departments). In one team where the external PO lacked knowledge about Agile, his engagement was improving as the time passed, and his knowledge about Agile was increasing. Therefore, the interviewees commented that their PO proxy may be removed since the external PO was reaching his adequate and satisfactory role fulfilling the team's needs and expectations.

“Product Owner is the key person for successful transformation, his knowledge about Agile is crucial, he represents the connection of the team to the outside world... when Product Owner is external, he is not so interested to participate in meetings and sometimes even does not see the value of Agile techniques or has proper knowledge about Agile framework, I mean this is applicable to all the stakeholders.” - team member, P12.

“Product Owner should be from our biggest client, for him it is impossible to participate in every daily meeting, therefore I am Product Owner and I communicate with him.”- Product Owner, P14.

4.2.6 High specialization and lack of multidisciplinary of team members

Technical expertise and skillset of each team member strongly influences the way of assigning new roles in the Agile setup. The interviewees pointed out that high specialization and a lack of multidisciplinary approach pose a problem regarding reorganization of teams. Their main concern was not to lose the specific technical skill by reallocating their member to the other team, and potentially create a bottleneck in their own team.

One team intended to split into three sub teams. Instead of reaching higher efficiency, the team members had to frequently exchange tasks within different sub teams due to specific expertise of each team member and sequential relations among tasks.

“Specialization is a problem to become truly Agile... We have only one business analyst in team and it represents a bottle neck, others should take those tasks or at least learn to do it to be able to take those tasks, we are lacking multidisciplinary people... sometimes we take tasks from product backlog instead of finishing all in sprint because of difference in expertise” - Product Owner - P14.

4.3 Consequences

We have identified four consequences in relation to the Core category: *delegation instead of collaboration, unbalanced workload, divided business and development and unproductive Agile ceremonies*. This section reviews each of these consequences.

4.3.1 Delegation instead of self-organization

Agile software development is based on 12 fundamental principles according to the Agile Manifesto (Fowler and Highsmith 2001) and we refer to two of them: *(1) projects are built around motivated individuals, who should be trusted and (2) self-organizing teams.*

Team members from one team pointed out that when there is lack of knowledge about Agile, or lack of trust between the Product Owner and the team members, the intrusion of the Product Owner in the team organization begins. Similarly, the Product Owner expressed similar issues in his case with stakeholders, or higher management representatives. If knowledge about Agile was low, higher management representatives and stakeholders intended to “micromanage” and interfere in the activities of the Product Owner. Self-organization was not respected and they were pursuing the “task delegation” management style.

According to the team members, they have the responsibility to implement the tasks in the sprint, and the Product Owner is responsible for the prioritization of the product backlog in accordance with the client’s demands. Any intrusion of the Product Owner on their autonomy was perceived as negative.

“In the sprint planning meeting, we do not dedicate enough time to elaborate tasks, mostly it is a moment where Product Owner delegates tasks....acts as a project manager” Scrum Master P11.

“Our Scrum Master does not want us to lose time in estimating points....moreover she is the first one to choose tasks and she takes too many for herself...she acts and as a Product Owner also and then she cannot fulfil the regular tasks” team member P17.

4.3.2 Unbalanced workload

Scrum Master should enable conditions for team members (S. Rubin 2013) and he or she has a crucial role in identifying a problem and involving new team members to increase the skill set in the team if needed. According to the comments of interviewees, the unbalanced workload was a consequence of the Scrum Master’s “traditional” management style.

Among team members, the role of the Scrum Master was perceived as an additional burden. The person acting as the Scrum Master would have additional management activities and increased workload without specific reward. Therefore, the interest to become Scrum Master was low, but surprisingly, all the interviewed Scrum Masters were satisfied with their role in the team.

Members of one team had the impression of unbalanced workload among the team members. The Scrum Master of the team grouped the tasks by the team member skill set. In that way she limited the team members to self-organize. The feeling of the team members was that the Scrum Master was delegating tasks, and consequently overloading certain resources in the team by preselecting the task types. In their opinion, the potential of the team was not used, and the Scrum Master herself was overloaded with the tasks that were not “given” to other available team members.

“The potential of the team is not used to the fullest extent. After Agile transformation process we noticed that work is not equally distributed among team members, it became visible once that one colleague had done almost double more points than the other one... Also, Scrum Master takes too many analysis tasks and others in the team can do it so it

should be distributed and others should learn and practice also... bottle necks are created due to distribution of tasks in a wrong manner.” - team member, P16.

“We have marked the tasks which are for business analysts and developers should not take them on. We have seen in last sprints that we are missing more business analysts or developers should take on those tasks, too, because we had to take on new tasks from product backlog, but we have not finished all the tasks in the sprint. We should have more business analysts in the team.” - Scrum Master, P18.

4.3.3 Divided business and development

In the research, Product Owner background proved to be an important factor in business and development integration or division. One of the fundamental principles (Fowler and Highsmith 2001) of the Agile software development is: *close, daily cooperation between business people and developers*. Since Product Owner prioritizes and gathers requests from all the stakeholders, he or she represents the business aspect of the project (S. Rubin 2013). The relation of the project owner with the development team implies integration of the business and development side of the project, going from divided to fully integrated.

Three out of five Product Owners were appointed from the business side, and the other two were appointed from the development side. Depending on their previous position in the company and their background, the relation of the Product Owner with the team was more or less intense. Integration of business and development was better if the assigned Product Owner had a business background. Having one business representative in the team created real synergy of both sides. In the five teams observed, intensive collaboration and communication of the Product Owner and the team was relatively fast to obtain, but if the Product Owner was assigned from the development side, it seems that the true integration of business and development was reached at a slower pace.

“By implementing Agile, I think that they wanted to connect more the business and development areas in the company with this change... Now we all have the common goal to improve the product.” - team member, P13.

“The product vision is more global and easier to transfer.”- Product Owners, P1, P6, P14.

4.3.4 Unproductive Agile ceremonies

Establishing and maintaining all the Agile ceremonies was identified as a challenge by the interviewees. The following meetings are considered Agile ceremonies in Scrum (S. Rubin 2013): daily planning, sprint planning, sprint review and sprint retrospective. The Scrum Master has the responsibility to enable fruitful environment to establish and maintain the ceremonies encapsulated in the Scrum method.

Different concerns were expressed by the employees regarding the Scrum ceremony frequency, participation and productivity. In the sprint planning ceremony, the unproductivity was triggered by different factors such as: use of different point scales for task estimation and interference of the Product Owner. In the sprint review ceremony, the identified obstacle to productivity was the low engagement and participation of the stakeholders. One team regarded the Agile retrospective meeting held at the end of sprint as unproductive. In their opinion, by that time the problems and potential improvements are

forgotten. Therefore, they had a short retrospective ceremony after the daily meeting if needed.

“I would like to try some new functionalities in the review... clients should be more involved and we need to try some user case... observing the report is not interesting and it is waste of time, you can just send it by mail.” - team member, P5.

“Team is not involved in review, only client and PO, clients not that interested and more checklist meeting.” - team member, P4.

4.4 Contingencies

Contingencies define strategies of practitioners for moderating: the causes influencing the core category and consequences arising from the core category. Therefore, the main goal of the enlisted contingencies is to moderate the relation of the causes and consequences to the core category - Inadequate transition of organizational roles. Five different contingencies, which are discussed in this section, have been identified: *Product Owner proxy, critical task distinction, management involvement in Agile Coaching, Scrum roles rotation and team reorganization.*

4.4.1 Product Owner proxy

One of the obligations of the Product Owner is to be in contact with clients, users and other stakeholders. Whenever the client was external, an internal resource was assigned in the company to perform the role of the Product Owner proxy. In the case company, three teams had external clients, so in each team one member was assigned to act as a Product Owner proxy.

In one team the key users were from different departments. The Product Owner was committed from the internal resources to act as a unique point for recollecting and analysing the requirements and improve the general communication in the project.

In the other team, the external Product Owner was not as responsive as expected and needed by the project team. As stated by the team members, the Product Owner was lacking knowledge about the Scrum method and an internal Product Owner proxy was appointed to the project. According to the opinion of the Scrum Master and team members, they were reaching the point when the Product Owner proxy was not needed any more, since the external Product Owner improved his knowledge and understanding of Agile methods and should be capable of fulfilling his role as expected. In this case, the Product Owner proxy was educating the external Product Owner on Agile methods and, at the same time, filled in the lacking communication between the Product Owner and the team, and activities that should have been performed by the Product Owner.

“Product Owner should be somebody from our client company but it is not the case. They are not interested to participate in daily meetings and other Scrum meetings, even though they like that we use Scrum. In the beginning we had one department as a client but with trust we are in contact with many of them and now it is hard to catch all the queries...Finally, we have appointed one person to be Product Owner from our side since otherwise communication would be chaotic and stakeholders would be complicated to manage.” - Product Owner, P14.

4.4.2 Critical task distinction

Specialization of team members or skills needed for specific tasks narrows down the possibility of the team self-organization. *Self-organizing teams* is one of 12 Agile principles (Fowler and Highsmith 2001).

One team included the critical task field in the task details to foster individual self-organization of the team members. The tasks were now divided into two groups: critical and non-critical. When a task was marked as non-critical, the team members were encouraged, free to choose and individually select tasks (self-organize), while in the case of critical tasks, all team members would discuss in the sprint planning meeting who would be the best to implement the task.

“We are a big team and some tasks simply have to be done by only one specific person, it is impossible to give it to anybody else. Our product is critical for the platform and we cannot send anything to production. We have some tasks that are not critical and we mark them and in this manner we let other team members to take them and improve their skills... in this way we let people choose to do what they want but only when it is possible.” - Product Owner, P18.

4.4.3 Management participating in Agile Coaching sessions

The participants in the education prior to the Agile transformation process were only team members and future Product Owners as middle management representatives. The higher management of the company was not involved in the education. Each team participated in one coaching session per month, lasting 20 minutes approximately. In the same day, after team coaching sessions, Scrum Masters would be invited to a cross team coaching session where they would, together with the Agile Coach, share their best practices. According to the interviewees, the coaching sessions were very helpful for the Scrum method adoption and fine tuning of the processes in the company. The additional benefit of the coaching sessions was to engage more the higher management in the Agile transformation process. Moreover, occasional participation of the higher management (in three sessions out of five) in the coaching sessions demonstrated their support. In this way the engagement of the higher management increased. Also, their knowledge about Agile and awareness about the real problems encountered by their employees in the process was better.

“We have invited higher management to coaching sessions, in one way to inform them about the changes that will happen and to manage their expectations. Additionally, their participation was important for the members of the team, to demonstrate their support and understanding of the Agile transformation process.” - Agile Coach, P21.

4.4.4 Scrum Master role rotation

According to Rubin, the essential Scrum roles are: Product Owner, Scrum Master and development team (S. Rubin 2013). Two teams changed their Scrum Master during the data gathering activities (six-month period).

One team had a problem of low authority of the Scrum Master. On the one hand, the Scrum Master was the newest member in the team, and on the other hand, the Product Owner had been working many years in the company, having good informal connections with the team members. Several times it was mentioned by the interviewees from different teams that the

clear benefit of being a Scrum Master was not obvious. Additionally, they had the feeling that it brought additional unrewarded work to the person dedicated the role of the Scrum Master. The researchers gained the impression that it was due to this reason that the youngest team member was appointed this role. The Product Owner continued in the old fashion “task delegating” style and the Scrum Master failed to create the culture of a self-organizing team. Finally, the Scrum Master was replaced after 4 months by another team member, and progress in team motivation was evident.

The other example of the regular Scrum Master rotation was identified in two teams. The objective of Scrum Master rotation was to balance the workload of team members. Since the Scrum Master role is perceived to add workload on the members, the team members agreed to change the Scrum Master among themselves every three or four months. To the researchers, the team members’ knowledge seemed to be on a similar level since all the members attended the Agile courses.

“Nobody wants to be Scrum Master, it is not paid extra and has extra work... we did not implement any reduction of programming hours for Scrum Master which would be a motivation... nevertheless I like the role of Scrum Master, it makes me feel good to see that the team is working well.” - Scrum Master, P7.

4.4.5 Splitting a large team

According to the interview respondents, the middle management tend to maintain the number of team members throughout the Agile transformation process. According to Rubin, teams bigger than 10 members should be divided and reorganized (S. Rubin 2013).

The largest team that participated in the research study had 16 team members. The team under study was not reorganized in the beginning of the Agile transformation. After five months of the Scrum method adoption, they showed the initiative to split the development team into two sub teams. In the new setup, the Product Owner and the Scrum Master participated in the meetings of both development teams. It was a recent change and the initial feedback obtained through observation activities, after two implemented sprints, was very good in the opinion of the researchers.

“We are big a team and since we are working on different functionalities of the same platform, we should be together... over 10 people is hard to manage and we are 16... meetings were not interactive and they did not provide value as expected. We have decided to split the team in two, but some of us attend meetings of both groups: Product Owner, Scrum Master and testers...in this way it works very well and it is a kind of project program management.” - Scrum Master, P19.

4.5 Contingency moderation of causes and consequences

In this section, relations among causes, consequences and contingencies will be further discussed. The section is based on the results obtained in the research and further analysis conducted by the researchers. The data presented in table 3 (causes, consequences and contingencies) are taken from Figure 3. Further on, the corresponding section number in the Manuscript of each cause, consequence and contingency is given in brackets. The moderation of contingency (positive or negative) on the appropriate cause or consequence is demonstrated in the cells of Table 3.

Table 3. List of contingencies and their positive/negative relation to the causes and consequences (with manuscript section number)

Contingencies	Causes		Consequences
	Increase	Decrease	Decrease
Product Owner proxy (4.4.1)		Lack of Product Owner engagement (4.2.5)	Delegation instead of self-organization (4.3.1) Divided business and development (4.3.3) Unproductive Agile ceremonies (4.3.4)
Critical task distinction (4.4.2)			Delegation instead of self-organization (4.3.1) Unbalanced workload (4.3.2)
Management participating in Agile Coaching sessions (4.4.3)		Lack of higher management commitment and support (4.2.3)	Divided business and development (4.3.3)
Scrum Master role rotation (4.4.4)		Strong team culture hinders changes (4.2.1) Lack of knowledge about Agile (4.2.2)	Delegation instead of self-organization (4.3.1) Unbalanced workload (4.3.2) Unproductive Agile ceremonies (4.3.4)
Splitting a large team (4.4.5)	High specialization and lack of multidisciplinary of team members (4.2.6)	Strong team culture hinders changes (4.2.1) Maintaining the original team size (4.2.4)	Unbalanced workload (4.3.2) Unproductive Agile ceremonies (4.3.4)

The detected relations among contingencies and causes and consequences are next described:

Product Owner proxy is moderating one cause and three consequences. The existence and the appointment of a Product Owner proxy was an effective strategy in the company to decrease the effect of the lack of Product Owner engagement. The Internal Product Owner (Product Owner proxy) improved communication in the case of having more users or in the situation of having an unengaged client. Also, Product Owner proxy decreased the division of business and development sides of the company as a key role in between. In one team, the extensive influence of the Product Owner “delegation” style was decreased by the Product Owner proxy. In one case, the existence of proxy Product Owner improved the Agile review meeting (Agile ceremony) and the meeting was considered more effective by the team members.

Critical task distinction is moderating two consequences. In one team, they proposed including additional notation on tasks. This technique improved self-organization, and workload of team members was more balanced. The team noted non-critical tasks which provided the opportunity for lower skilled employees to self-organize and to choose freely among those tasks and, finally, to be equally loaded with work as the other members.

Management participating in Agile Coaching sessions is moderating one cause and one consequence. The participation of higher management in the coaching sessions demonstrated their commitment to the Agile transformation process, the understanding of current issues and challenges faced by team members and, as a result, increased their knowledge about Agile methods. Also, it positively influenced the integration of business and development. One coaching session brought discussion about double reporting to higher management. Team members expressed possible simplifications since Scrum artefacts could be used as part of the reporting procedures.

Scrum Master role rotation is moderating two causes and three consequences. Scrum Master rotation positively influences the transfer of knowledge about Agile, decreases the influence of strong team culture and opens a team member to be more prone to changes. Team members recognized the Scrum Master role as a necessary role, but as an additional workload to the person performing this role. Therefore, in two teams they agreed to change the Scrum Master among themselves on a regular basis. Team members who were to become Scrum Master were motivated to learn more about Scrum. As it was recognized as an additional workload, Scrum Master rotation balanced the workload on team members. In the case of having a Scrum Master who was still in the traditional project manager mindset, rotation helped to decrease delegation and to improve Agile ceremonies such as sprint planning.

Splitting a large team is moderating three causes and two consequences. For mature, well established teams, structural reorganization “awakes” the team and has a positive effect on the team culture, and by team division a new culture which is more positive to changes can be established in teams. When teams are divided, certain skills may be missing in a team, thus influencing the skill development of the team members. On the contrary, reorganization might cause the inflow of new skilled resources, thus positively influencing the knowledge sharing among team members. In two teams, the researchers recognized the resistance to team reorganization which seemed to be common, especially among mature teams. One team, due to a large number of team members, decided to divide and reorganize. Team members from the biggest team in the research (16 members) commented about unproductive Agile ceremonies such as daily meetings and spring planning meetings. Also, due to the large number of team members, they had an impression that the Scrum Master and the Product Owner were overloaded. Reorganization of the team had a positive effect on the identified consequences.

5. Discussion

In this section, we discuss and relate our findings to the research reported by others. The core category identified in the research is the inadequate transition of organizational roles. The core category reflects the main challenge identified in the Agile transformation process in the company where the research was conducted. Therefore, we refer to the literature and research on organizational roles in software development and effects of the organizational role transition from traditional towards Agile. Further, we discuss organizational roles defined both in the traditional framework and in Agile frameworks.

It was identified that the situational context of each team participating in the research was different and the employees had different strategies to handle the issues and challenges arising during the transformation. Software development process is dependent on the situation characteristics of individual software development settings (Clarke and O'Connor 2012). In their research, Clarke and O'Connor classified 44 factors and 170 sub-factors affecting the software development process. In our research, different causes, contingencies and consequences can be directly related to some of the proposed situational factors

affecting software development such as: management commitment, management expertise, team size, team culture, technical skill, individual knowledge and engagement. Each situational context presents its own unique set of challenges and before defining the meaning and the scope of each organizational role, the organization should analyse the situational factors of each team (Clarke and O'Connor 2015).

The Scrum method proposes the following organizational roles: management, Product Owner, clients, users, Scrum Master and team member (Schwaber and Beedle 2001; S. Rubin 2013). Each Agile method proposes a different set of organizational roles. The research of Yilmaz et al. provides a good comparison of roles in the following software development approaches: plan driven, ISO/IEC 12207, system engineering, Scrum, extreme programming and feature driven development (Yilmaz, O'Connor, and Clarke 2012a). In the company, Organizational roles are well defined for each of the approaches mentioned, but the question how we should “transition” or map the organizational roles from one to another approach remains unclear.

Terminology used in different approaches might also bring additional confusion in the process of Agile transformation. In terms of terminology, it can be discussed if an organization is using an Agile methodology or method or processes or ecosystem (Clarke et al. 2016). Similarly, there seems to be significant overlapping of some terms like for instance, *Scrum Master* and *sprints* compared to *project/process manager* and *iterations*. There is an affinity to create new titles of software development roles. Scrum Master is not named project manager in order to emphasize the drastic change managers must make in their approach (Schwaber 2004). The Scrum Master is indirect and he should be the knowledge distributor and he should be policing Scrum practices as a limit of their power (S. Rubin 2013). This role could, to a certain extent, refer to the process manager in the traditional organization. If Scrum Master was a project manager, he would naturally tend to assume his authority as a traditional project manager (Moe, Dingsøyr, and Dybå 2009). In our research we have identified similar issues of Product Owner or Scrum Master interference in regular team activities. The consequences of this phenomenon are embedded in the resulting theory as delegation instead of self-organization and unproductive Agile ceremonies.

Agile methods have well established the definition of practices, roles, ceremonies and artefacts. The question is the overlapping of differently named roles and, more importantly, how should the organization tailor and transition the roles to the new method. Selection of the proper method to be adopted is not enough for achieving goals of software organizations. The organizational roles should be tailored to an individual software project depending on the development activities (Yilmaz, O'Connor, and Clarke 2012b).

In the following paragraphs we discuss our findings in relation to each organizational role in the Scrum method:

(1) *Management*. The Agile transformation process in the observed company had started as a top-down approach. The Scrum method was chosen by the company management as an Agile practice to be used in the company, and it was firstly implemented at the level of development teams. Development teams were the first to go through the education cycle about Agile practices. The transformation at the team level affects the organization as a whole, other departments and higher management structures (Cohn and Ford 2003; Van Waardenburg and Van Vliet 2013; Lindvall et al. 2004b). At first, the Agile transformation process is confusing for both developers and management (Van Waardenburg and Van Vliet 2013).

The results of this research study have shown that the lack of higher management commitment and support, and their knowledge about Agile influenced the inadequate transition of organizational roles. Participation of higher management in the Agile

Coaching sessions was an effective strategy to increase their knowledge and to demonstrate their commitment to the Agile transformation. It takes time and effort to change the attitude we-them (Boehm & Turner, 2003; Van Waardenburg & Van Vliet, 2013). Implementation of Agile practices improves one part of software development process - project level, but leads to new issues of the other part - management level (Petersen and Wohlin 2009).

Creating double reports for higher management was pointed out by the interviewees as a problem. A lot of information could be obtained directly from Scrum artefacts. Therefore, some reporting habits connected to traditional software development were not needed anymore. Traditional processes appeal to many in upper management since they facilitate progress tracking (Cohn and Ford 2003) and they might be resistant to change towards Agile since the reporting system has to change, as well.

(2) *Product Owner* is a key person for successful integration of business and development sides (S. Rubin 2013). New responsibilities and constant engagement of the Product Owner, compared to the traditional role of the product manager or project manager, implies significant changes in the management culture, and devotion of more time to interaction and integration of development with stakeholders. The Product Owner is the empowered central point of product leadership and he or she has a set of different responsibilities enabling him or her to: manage economics, participate in planning, perform product backlog grooming, define and verify acceptance criteria, and collaborate with development team and stakeholders (S. Rubin 2013; Sutherland 2014). The Product Owner is also considered to have a “single wringable neck” position of his responsibility of prioritizing the product backlog.

Business representatives are used to participating in planning in terms of defining requirements at the beginning, defining end dates, delegating tasks and providing feedback when development process is finished (Hoda, Noble, and Marshall 2011; Van Waardenburg and Van Vliet 2013; B. W. Boehm et al. 1987). The research conducted showed that lack of proper Product Owner engagement influences the inadequate transition of organizational roles. Two types of interferences of the Product Owner in the team dynamics were observed: excessive commanding style or lack of presence and low interest. According to the results in the company, the appointment of the Product Owner proxy decreased the lack of Product Owner engagement.

Management should change the style from delegation towards collaboration (Nerur, Mahapatra, and Mangalaraj 2005), resulting in the integration of business and development sides of the organization. Divided business and development was identified in the research as one of the consequences of the core category. This phenomenon is primarily the responsibility of the Product Owner acting as the “connection” of business and development sides. When the assigned Product Owner had more development experience, it was more likely that he or she would tend to perform and “micromanage” (Cohn and Ford 2003) the development activities of the team. On the other hand, if the Product Owner had more business experience, involvement in daily activities and different Scrum meetings was on a lower level. Based on the obtained data in the company, background and previous experience were important parameters affecting the way the Product Owner would perform the newly assigned role in the Scrum framework.

The traditional product manager role has served many companies well for many years and now it would be the closest to the Product Owner role (Fraser et al. 2004). High level decisions about a system under development would have been expected from the product manager, but the Product Owner should be familiar with the operation of business processes and the division of technical work in small packages. This statement raises a question from the beginning about the role transition and who should become the Product

Owner: project manager, product manager, system analyst or someone else? Should the pattern be defined at the organizational level or should it be team specific?

In the case company, five Product Owners were interviewed. In Table 1 their role before Agile adoption is shown. Three Product Owners were project managers and two of them were system analysts in the previous organizational setup. According to the interviewees' responses, two out of three teams where project managers were assigned the Product Owner role, experienced problems with the Product Owner authority and their "micromanagement". The third team where the project manager was assigned as the Product Owner had a problem with the number of team members, and it was reduced at a later stage. None of the interviewees stated clearly the reason for having 16 members in the team, but the impression of the researchers was that the high authority of the Product Owner (previously project manager) influenced and delayed the decision to split the team into two. The cohesion of the other two teams where system analysts were assigned the role of the Product Owner did not experience problems with the authority of the Product Owner. In addition, these two teams were the ones to establish regular Scrum Master rotation. In one of those two teams, the interviewees stated that the requirement checklist would be prepared by the team for the Product Owner. It was mentioned that the Product Owner sometimes fails to ask stakeholders enough about product details and business aspects in general, and this list should be of help as a reminder. In the opinion of the researchers, Product Owners coming from the technical side (system analysts) were a bit limited with soft skills, but the cohesion of the teams was much better from the beginning. It should be kept in mind that the majority of the interviewees were from the technical side of the company and they could be inclined to some extent towards the opinion of technical staff taking the management roles. Therefore, based on the case findings it can be concluded that Product Owners assigned from the technical side tend to influence the team on HOW to do things and do not focus on WHAT should be done and they tend to be missing soft skills. On the other hand, if the Product Owner is assigned from the business side, they tend to delegate and have the traditional managerial style. Business side Product Owner tends to integrate the business and the development side faster, while the technical Product Owner tends to establish team cohesion much faster. In the case company, a Product Owner proxy was assigned for each external Product Owner, which happens to be a good strategy to start the transformation of the team and, later on, to exclude the Product Owner proxy if possible. According to the results, the Product Owner should be assigned from the business side while the Product Owner proxy should come from the technical side.

(3) *Stakeholders (clients and users)*. Different obstacles could pose a problem to the expected client involvement: knowledge about Agile values (Gandomani et al. 2015), too many end users/clients involved in the product (Fraser et al. 2004) or geographical location and general availability to join meetings (Grisham and Perry 2005). Having more than one customer may cause problems (Fraser et al. 2004) and a solution would be to identify the key customer who would further manage other customers (stakeholders). Customer collaboration over contract negotiation is one of the fundamental values stated in the Agile Manifesto (Fowler and Highsmith 2001). Customer involvement and satisfaction (Soundararajan and Arthur 2011) is one of the benefits of the successfully adopted Agile framework. Extreme programming brings the customer and development team together into a tight functional unit, but there is a danger that effective Agile methods might weaken the customer relationship (Grisham and Perry 2005). In most of the cases, a single person should be and is able to perform the Product Owner role, but under different circumstances Product Owner proxies may be of help (S. Rubin 2013). As identified in the research, the Product Owner proxy existence was an efficient contingency to moderate the lack of Product Owner (stakeholder) engagement. Communication with clients was improved and the requirement process was better. Also, the Product Owner proxy may have an important role of educating stakeholders about the Agile practices used by team members.

(4) *Scrum Master and team members*. Scrum Master is a new role when compared to traditional software development. In Scrum, many project management tasks are taken over by project teams (Stettina and Hörz 2014). The Product Owner is focused on creating the right product, the development team on constructing the product in the right way and, the Scrum Master is focused on helping everyone to embrace Scrum values, practices and principles (S. Rubin 2013; Sutherland 2014).

Regular Scrum Master rotation among team members was identified in two company's project teams. Team members were appointed the Scrum Master role for a limited period of time. Most of the interviewees recognized the Scrum Master activities as an addition to daily development activities. Typical situations in Scrum teams when they should rotate the Scrum Master are: when team members struggle to understand the duties of the Scrum Master, more team candidates with similar level of knowledge could become Scrum Master, to establish balance of power among team members and make a shared team responsibility in the team (Cohn 2012; Schatz and Abdelshafi 2006). In the case company, in teams with the internal Product Owners, who were also having previous technical background, it was decided to establish regular Scrum rotation. The interviewees were positive about this practice and the main reason to establish such behaviour in the opinion of the researchers was relatively equal knowledge on Scrum practices and perception of additional workload of being a Scrum Master.

One of the solutions suggested by team members was to establish a reward system for the additional effort of the Scrum Master. Another suggestion was to decrease the workload of the Scrum Master. Rewards systems should be reoriented to recognize both individual and team contributions (B. Boehm and Turner 2005). Unbalanced workload is embedded in a resulting theory in Figure 3 as consequence and Scrum roles rotation was defined as an effective contingency identified in the company.

The existence of self-organizing teams is one of the principles stated in the Agile Manifesto (Fowler and Highsmith 2001). In traditional software development, role assignment is individual and favours specialization. In Agile software development, self-organizing teams and role interchangeability are encouraged (Nerur, Mahapatra, and Mangalaraj 2005). Team members have commented in the research that excessive interference of the Scrum Master or the Product Owner in task distribution among team members is a common problem at the beginning of the Agile transformation and that it limits self-organization.

6. Evaluating the results

There are diverse perspectives and approaches to evaluating a grounded theory research study (Birks and Mills 2012; Angen 2000). We used Glaser's approach (B. G. Glaser 1978) and Lincoln and Guba's criteria (Lincoln and Guba 1985) to evaluate the research process and to validate the emerged theory. These made us confident of the trustworthiness of the findings.

6.1. Glaser's approach

Glaser and Strauss consider that GT is a good method for building theory, not verifying it. Validity in its traditional sense is not a critical aspect in GT (B. G. Glaser and Strauss 1967). Therefore, follow-up studies to verify the emergent theory are left to other research methods that are better suited to that purpose (B. G. Glaser 1978). Glaser believes that an

emerged GT is grounded in data and in some sense has already been verified: “A grounded theory gets its concepts from the data; it does not bring ideas to force the data that need to be subsequently tested” (B. G. Glaser 1978). Glaser proposes four criteria for evaluating the credibility of the emergent theory (B. G. Glaser 1978): *fit*, *workability*, *relevance*, and *modifiability*.

Fit criterion mainly focuses on emerging codes, categories and theory from data rather than researchers’ preconceived perceptions and biases. Researchers must remain open and sensitive to data. The GT should fit and explain the experiences of the other practitioners that do not participate in the theory’s generation process. *Workability* is used to assess the integration of the core category and its main related categories. On the basis of workability, a theory should be able to explain what is happening in the area under study and predict what will happen. Furthermore, it should describe how the major concerns of the participants are continually resolved. *Relevance* refers to how well a GT focuses on the main problems and processes that emerge in a substantive area. (B. G. Glaser 1978) believes that “GT arrives at relevance, because it allows core problems and processes to emerge”. *Modifiability* focuses on the ability of the theory to be continually modified upon receipt of new data. The emerged theory must be changeable in such a way that it maintains its validity and also fits with the data. A category, which fits and works, is relevant and is subject to continual modification and thus, will integrate a theory in the way that the theory is dense and saturated with relationships (B. G. Glaser 1978).

In this research, we avoided conducting extensive literature review at the initial stage of the study. Knowledge of the impact of Agile roles definition was primarily acquired through interaction with the research participants while collecting data from them. Moreover, we employed a theoretical sampling technique, in which the decision about the next data (what and where) was made based on the emerging theory. We tried to recruit appropriate participants in different roles across different teams to contribute most relevant explanations on the emerging concepts and from diverse perspectives. Furthermore, we followed Glaser’s recommendation regarding the interviews questions and continually changed them in order to solely focus on the concerns that seem central to the emerging concepts, categories and theory (B. G. Glaser 1978). Theoretical sampling was continued until theoretical saturation was noticed in the new data. All the concepts, categories and their properties which together define the grounded theory only emerged from substantive data without being forced by preconceived hypotheses or ideas. Consequently, we believe that the emerged theory and its related categories *fit* the realities under study.

One way to assess *workability* and *relevance* criteria of the theory is to ask the participants whether the core category and its related categories relate to the substantial problems in the topic under study. To do this, while data analysis was ongoing and concepts were emerging, we attempted to write different technical reports explaining the emerging concepts and theory. Then, we provided those participants who were interested with the reports and asked them to give feedback on the emerging categories. Moreover, the researchers presented in a regional symposium the different aspects of the emerging theory to the Agile community at regional level. Fortunately, we received good feedback from various practitioners that reflect on how the emerged theory is meaningful, useful, and applicable to them. According to the received feedbacks, we argue that the theory not only works but is also sufficiently relevant to the subject under study.

The emerging theory was *modifiable* during the research study. The theory evolved over time when we compared new data to the existing concepts and categories. Agile roles definition, their principle characteristics, the transition challenges, facilitators, and prerequisites evolved and matured through the research as we went discussing with different participants. The researchers strongly believe that the emerged theory can still

continue to evolve by receiving new relevant data and comparing these with the existing theory. Therefore, we argue that the findings are readily modifiable.

6.2. Lincoln and Guba's approach

Lincoln and Guba suggest some criteria to evaluate a qualitative research including credibility, dependability/auditability, confirmability, and transferability (Lincoln and Guba 1985). As they state, credibility is the most important criterion in establishing trustworthiness. Credibility is achieved through *peer debriefing*, *triangulation of data*, and *member checking* (Lincoln and Guba 1985). The rest of this section explains how this study was evaluated against the credibility criterion.

Peer debriefing: Credibility was incorporated into this research with the use of peer debriefers. The primary researcher continually discussed the GT process and the emerging concepts with the other authors and the Agile Coach who was responsible for the Agile transformation process, with the main goal of getting feedback to ensure validity. During this stage of the research, each of the full interviews was provided to the peer debriefers to read the text line-by-line, in the same way as the primary researcher, and highlight relevant issues or concepts. After independent reading of the interview, the peer debriefers were provided with the codes and memos discovered by the primary researcher. Peer debriefers questioned the primary researcher about his observations and helped to clarify interpretations, contributing to the credibility of the emerging theory.

Triangulation of data: We conducted the initial interviews primarily to collect data from the research participants. Then, we applied theoretical sampling and interviewed several participants from different roles including Product Owners, Scrum Masters and team members (with different levels of experience) in order to obtain a comprehensive view of the research phenomenon under study. Moreover, as literature is perceived as a great source of data, we carried out a major literature review after grounding the emerged concepts in order to integrate them into the theory. Consequently, the data triangulation procedure was used automatically during this study.

Member checking: Although all the 21 interviews were voice recorded, we transcribed them for ease of use during the data analysis stage. Different measures were adopted in order to verify the data collected from the participants. For example, we provided the participants with their respective interview transcriptions and we requested and encouraged them to verify the data provided, suggest corrections on wrong interpretations and provide any further information. Moreover, they were also provided with the report with the results to be presented in the regional Agile symposium in order to get feedback on the emerging theory. We treated the received feedback as another source of data that was integrated into the emerging theory through the data analysis process.

7. Limitations

Since all of the emerged codes, concepts, and categories came from the data directly collected from the context, the results and findings of this study are grounded in real environments (B. G. Glaser 1978; B. G. Glaser and Holton 2004). A theory produced by grounded theory research cannot be claimed to be universally applicable. Nevertheless, this research does not state its results are universal because the study was limited to those volunteers who accepted participating in this research. However, it claims that its findings characterize and depict the area under consideration, but the inherent limitation of GT is

that the emerged theory is only grounded in the particular context investigated in the research (Adolph, Hall, and Kruchten 2008).

Regarding the participant selection procedure, we communicated our requirements and needs from participants to the company. Many members were willing to participate, so we selected some members of each work team to be interviewed. This decision was made considering the role of each member, trying to have at least one representative of each role in the team. We interviewed participants from the Agile teams, and not business representatives. Consequently, the issues identified may be inclined towards the Agile perspective.

GT's theoretical sampling prescribes the researchers to analyse and code an interview before carrying out the next one. This was not possible in our study, as we were asked to conduct diverse interviews in the same day. The second interviewer was only able to transcribe the most important parts of the responses in real time. After an interview, the researchers were able to add some more important quotes in the transcription. While the final coding was not conducted until all the interviews had been completed, during the successive interviews, the emerged concepts and issues deepened in the earlier interviews were taken into consideration.

Our study took place in one organization only. We are planning to widen the sample of participants in other companies with similar characteristics in terms of the Agile transformation process, but we have not done a similar study yet.

8. Conclusion and future work

This article presents a grounded theory study of Agile transition involving 21 Agile practitioners from a large software development organization in Spain. The theory demonstrates how enterprise factors influence the transition of organizational roles and how practitioners cope with the challenges during the Agile transformation process.

The data collection process was done through interviews and participation in coaching sessions and observation activities. Participation in coaching sessions and observation activities proved to be very useful for gaining insight in team dynamics and their daily activities.

This research study helped the company's personnel to identify potential improvements and better observe the current situation. Another round of interviews is planned in a year's time to measure the progress of Agile transformation and obtain new inputs.

One of the contributions of this paper to the research community is a detailed explanation of grounded theory application in the Engineering field. GT was an adequate method for the research conducted. It was very useful for grasping human interaction at the lowest level, since Agile software development is human centric. During the interviews it was interesting to observe different points of view on certain issues, coming from different members of the same team.

The results of this study suggest more implications for the industry. Organization of courses followed by Agile Coaching sessions was an adequate way to distribute the knowledge about the Scrum method and start the Agile transformation in the company. Involvement of higher management in the Agile Coaching sessions showed good results in terms of demonstrating higher management commitment to the transformation and improvement of their knowledge.

The interviewees commented that the objective of the Agile adoption process was not transmitted clearly to team members at the beginning of the Agile transformation process. We recommend transmitting the objective of the transformation and expected results to employees involved at the beginning of the Agile method adoption.

Regular Scrum Master rotation was implemented in two teams and the interviewees commented on its positive influence on team cohesion and balancing the amount of work done by each team member. The Scrum Master role was perceived as an additional workload on team members performing the Scrum Master role. The teams established a system of Scrum Master rotation to distribute equally the workload of team members.

Team members proposed to establish a reward system to facilitate the Agile transformation process. The reward system could make the Scrum role more attractive, but also a more detailed system of rewards could facilitate the general process of Scrum method adoption.

Several topics worth further investigation were identified during this research study endeavour.

Scrum Master rotation suggested as a strategy to overcome inadequate role transition could be further investigated in other enterprises. It would be interesting to know if it is a common strategy in other Agile environments and the exact effect of this practice on team members.

We identified many Agile transformation frameworks in the research literature but we failed to find more details on how the organizational roles change during the transformation. Agile transformation frameworks describe factors influencing the changes, steps for implementing practices, but the transformation path of the organizational roles is not specified. It would be valuable to conduct a research which would provide potential strategies on organizational roles transformation. Also, identifying factors for organizational roles tailoring would be an interesting result for industry and research society. We consider the theory presented in Figure 3 as a small contribution to this topic, but we suggest further studies in this direction.

In the future it would be useful to perform similar research in more organizations in order to further refine the results presented in this research. Therefore we suggest similar studies should be performed in companies. Conducting similar research in other Companies would improve the covariance aspect which demonstrates dependencies between causes and consequences, which were not presented in this research due to the limited sample.

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Appendix A.1

Questionnaire adopted from the research (Paasivaara and Lassenius 2014) from the Appendix A. Interview questions A.1.

1. Transformation questions

- a) What is your role, background and experience at Idiso? What is your background in Agile framework and Agile method?
- b) What kind of training have you had on Lean and Agile software development? Would you have needed more training?

2. Agile and lean transformation

- a) Why was the transformation started?
- b) Explain the timeline of the transformation.
- c) Why did you choose Scrum?
- d) What comes from Scrum in your way-of-working?
- e) What are you still planning to change/improve?
- f) What is your general feeling of the transformation?
- g) Good/bad/challenges? How have you solved the challenges?
- h) What would you have done differently? How?
- i) Has Lean and Agile solved the challenges you wanted to solve by this change?
- j) How have you measured the change? What have you measured exactly? How does this change look according to your measurements?

3. Organization structure

- a) What does your organization look like? (draw a picture!)
- b) How has the Agile transformation affected the roles and responsibilities?
- c) What roles do you currently have in your organization?
- d) Are the following roles and their responsibilities clear: Project Manager vs. Scrum Master vs. Product Owner?
- e) Are the roles clear in your organization? Are there challenges/ improvement needs regarding these roles or their application?
- f) Comment on the organization structure: Good? Improvement needs? Do you still plan to change something regarding the organization structure?
- g) Do the organizational roles from before the transformation still exist in the company?
- h) How was it chosen who will become team Scrum Master and Product Owner?
- i) Did you identify certain overlapping of organizational roles or is it clear who should do what?
- j) Do you see the benefits of being assigned the role of Scrum Master and Product Owner?

4. Coaching

- a) External coaching: What kind of coaching/consulting have you had to support the transformation?
- b) What kind of training did you have?
- c) Have you had internal coaches? If yes, tell more about their role and tasks.

5. Team-level Scrum practices

- a) Your own team: team size, what does your team do, connections to other teams?
- b) Scrum practices of your team (explain all Scrum practices your team uses)

- c) What is the length of your iterations? What is your opinion on the iteration length? Explain.
 - d) Cross-functional teams: How has the idea of the cross functional teams worked out? Why? Why not?
 - e) Tell about the communication inside your team. Good/bad/improvement needs.
 - f) How and when do you communicate with other teams?
 - g) Do you know enough on what is happening in the other teams/elsewhere in the project? Is there something that you would need to know more about? What? Why?
 - h) What is working well in your team/regarding your team practices? What are the biggest problems? What should be improved? How?
6. Improvement suggestions
- a) Is there something that should be improved in your team?
 - b) How?
7. Tools:
- a) What are the most important tools that you use?
 - b) Tell a bit about each tool (what is it used for, who is using it, etc.) Good? Bad?
8. Scaling Agile/cross-team coordination practices
- a) What are your current (scaling) cross-team coordination practices?
 - b) Scrum-of-Scrums? Communities of Practice? Feature Owners? Tell about each practice.
 - c) How have you taken into account the global distribution in cross-team coordination? How do you handle communication? Tools used to support the communication and coordination? What kind of challenges have you experienced?
9. Measuring
- a) What do you measure?
 - b) How do you measure quality? Productivity? How has this changed compared to the way of working before the transformation?
 - c) What should be measured? Why?
10. Product management
- a) How is product management taken care of?
 - b) Where do you receive the requirements? Describe the whole flow from the customer request to the requirement being part of the product.
 - c) How is customer communication taken care of?
 - d) What does the Product Owner do?
 - e) Communication between the Product Owners/proxy Product Owners? Communication with the teams?
 - f) How is work divided between teams? Who does that? What are the current challenges of product management?
11. Opinions
- a) How do you think that the Agile transformation process has succeeded?
 - b) How do you think that Lean and Agile software development fits in your organization? What are the benefits this brings? What are the challenges? Has your opinion towards the Lean and Agile changed somehow during the transformation?

- c) What advice would you give others considering the initiation of the Agile transformation process or a similar action?
- d) What are your expectations towards our research?

***In the second round of interviews when the core category was identified the questionnaire was changed:**

- The questions that were removed: 2i), 4c), 5b), 8b)
- The questions that were added: 3g), 3h), 3i), 3j)

The interview length was 1 hour on average, the researchers made sure to spend more time with questions in the section 3 in the second phase of data gathering activities (12 minutes as a minimum).

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